Analysis of PDEs

Shing Tak Lam

Michaelmas 2023*

Contents

1 Basics **1**
1.1 Examples of PDEs . . . 1
1.2 Data and Well-Posedness . . . 3
1.3 Classifying PDEs (of order k) . . . 3

2 Cauchy-Kovalevskaya theorem **4**
2.1 ODE theory . . . 4
2.2 Real analyticity and majorants . . . 5

3 Cauchy-Kovalevskaya for PDEs **9**

4 Sobolev spaces **10**

5 Elliptic PDEs **10**

6 Hyperbolic PDEs **10**

Lecture 1

1 Basics

Let $\mathcal{U} \subseteq \mathbb{R}^n$ be an open set. A *PDE of order k* is an expression of the following form

$$F(x, u, Du, \dots, D^k u) = 0 \tag{1}$$

where $u : \mathcal{U} \to \mathbb{R}$ is the unknown, $F : \mathcal{U} \times \mathbb{R} \times \mathbb{R}^n \times \cdots \times \mathbb{R}^{n^k}$ is a general function. We say that u is a *classical solution of* eq. (1) if u satisfies eq. (1) in $\mathcal{U}$.

We may also consider the case where $u(x) \in \mathbb{R}^p$ and $F \in \mathbb{R}^q$, and we call this a *system of PDEs*.

1.1 Examples of PDEs

Example 1.1 (ODE system)
One example of a model used in mathematical biology is the ODE system

$$\frac{\mathrm{d}u}{\mathrm{d}t} = f(u, v)$$
$$\frac{\mathrm{d}v}{\mathrm{d}t} = g(u, v)$$

Example 1.2 (Laplace's equation)

*Based on lectures by Zoe Wyatt. Last updated October 10, 2023.

$$\Delta u = \sum_{i=1}^{n} \partial_i^2 u = 0$$

The Laplacian is an *averaging* operator.

Example 1.3 (Heat equation)

$$u_t = D\Delta u$$

This is also called the diffusion equation, D is called the diffusion constant.

Example 1.4 (Navier-Stokes)

The Navier-Stokes equations in fluid dynamics is

$$u_t = v\Delta u - u \cdot \nabla u - \nabla \rho + f$$
$$\operatorname{div}(u) = 0$$

Example 1.5 (Transport equation)

The transport equation is

$$u_t + v u_x = 0$$

where v is a constant, corresponding to the velocity. A modification is the *advection-diffusion* equation,

$$u_t + v \cdot \nabla u = D\Delta u + f$$

Example 1.6 (Poisson equation)

$$\Delta u = f$$

Describes electric field due to some charge, or Newtonian gravity.

Example 1.7 (Wave equation)

$$\Box u = -u_t t + c^2 \Delta u = 0$$

This models sound waves, seismic waves, ...

Example 1.8 (KdV equation)

This equation admits *soliton* solution.

$$u_t + \partial_x^3 u - 6u\partial_x u = 0$$

Example 1.9 (Maxwell equations)

$$\begin{aligned}\operatorname{div}(E) &= \rho \\ \operatorname{div}(B) &= 0 \\ \partial_t E &= \nabla \times B = J \\ \partial_t B &= -\nabla \times F\end{aligned}$$

Example 1.10 (Einstein's equations)

$$\operatorname{Ric}(g) - \frac{1}{2} g R(g) = 0$$

1.2 Data and Well-Posedness

All of the examples from above need additional information to solve, which we call the *data*. For example, we might need $u|_{\partial\mathcal{U}}$ and so on. A guiding principle to this process is called *well-posedness (in the sense of Hadamard)*.

We say that a PDE problem (equation and the data) is well-posed if we have

1. A solution exists (in some function space).
2. Given some data, the solution should be unique (depends on the function space of choice).
3. The solution depends continuously on the data.

The aim is to find the largest space for which a solution exists, but small enough so that it is unique. (For example, strong cosmic censorship in GR?)

Notation 1.11 (Multi-index notation)**.** We will use multi-index notation,

$$\mathbb{N} = \{0, 1, \dots\}$$

$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ is called a *multi-index*, and we define the *order* of α

$$|\alpha| = \alpha_1 + \alpha_2 + \cdots + \alpha_n$$

and the α-th derivative is

$$D^\alpha f(x) = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n} f$$

If $x = (x_1, \dots, x_n)$, then $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, and

$$\alpha! = \alpha_1! \cdots \alpha_n!$$

1.3 Classifying PDEs (of order k)

We say eq. (1) is *linear* if F is a linear function of x, u and its derivative. That is, we can write it as

$$\sum_{|\alpha| \le k} a_\alpha(x) D^\alpha u = f(x)$$

Moreover, we say that a linear PDE is *homogeneous* if $f = 0$. We say eq. (1) is *semilinear* if the highest order derivatives appear linearly with coefficients depending only on x. That is, we have

$$\sum_{|\alpha| = k} a_\alpha(x) D^\alpha u(x) + F(x, u, Du, \dots, D^{k-1}u) = 0$$

One example would be $\Delta u = u_x^2$. Finally, we say eq. (1) is *quasilinear* if the highest order derivatives appear linearly, but the coefficients depending on lower order derivatives. That is, we have

$$\sum_{|\alpha| = k} a_\alpha(x, u, \dots, D^{k-1}u) D^k u + F(x, u, \dots, D^{k-1}u) = 0$$

For example, we can have $uu_{xx} + u_{yy} - u_x^2 = 0$. Finally, we say eq. (1) is *fully non-linear* if it is non of the above.

2 Cauchy-Kovalevskaya theorem

2.1 ODE theory

Fix $\mathcal{U} \subseteq \mathbb{R}^n$ open, and suppose $f : \mathcal{U} \to \mathbb{R}^n$ is given. We would like to consider the ODE

$$\begin{cases} \dot{u}(t) = f(u(t)) \\ u(0) = u_0 \end{cases} \tag{2}$$

Theorem 2.1 (Picard-Lindelöf)**.** Suppose we have $r, K > 0$ such that $B_r(u_0) \subseteq U$ and

$$\|f(x) - f(y)\| \leq K\|x - y\|$$

for all $x, y \in B_r(u_0)$. Then there exists $\varepsilon > 0$, depending on K, r and a unique C^1 solution $u : (-\varepsilon, \varepsilon) \to \mathcal{U}$ solving eq. (2).

Lecture 2

Sketch proof, see examples sheet 1. If $u \in \mathbb{C}^1$ solves eq. (2), then by the fundamental theorem of calculus, u satisfies the *weak formulation*

$$u(t) = u_0 + \int_0^t f(u(s))\mathrm{d}s \tag{3}$$

Moreover, if $u \in C^0$ is a solution to eq. (3), then it is a C^1 solution to eq. (2). Moreover, if u exists, then it is a fixed point of

$$G(w) = u_0 + \int_0^t f(w(s))\mathrm{d}s$$

Let $\mathcal{S} = \left\{w : (-\varepsilon, \varepsilon) \to \overline{B_{r/2}(u_0)}\right\}$ continuous. We want to show that $\mathcal{S}$ is a complete metric space, $G : \mathcal{S} \to \mathcal{S}$ is a contraction for ε sufficiently small, and by the contraction mapping theorem G has a fixed point. □

Remark 2.2.

1. The solution (in general) can't be global. Consider for example
$$\dot{u}(t) = u(t)^2 \quad \text{with} \quad u(0) = u_0 > 0$$
Solutions to this equation blow up in finite time.
2. This does not apply to
$$\dot{u}(t) = \sqrt{u(t)} \quad \text{with} \quad u(0) = 0$$
There are two solutions. Note we *can* apply the Peano existence theorem.

Now suppose f is smooth, and we have $\dot{u}(t) = f(u(t))$ is C^1. By the chain rule,

$$\ddot{u}(t) = Df(u(t)) \cdot \dot{u}(t) = f_2(u(t), \dot{u}(t))$$

which is continuous. Hence $\ddot{u}$ is continuous, and so $u \in C^2$. Repeating this, we get that $u \in C^k$ for all k. That is, u is smooth.

In principle, given $u_0 = u(0)$, we can determine

$$u^{(k)}(0) = F_k\left(u, u', \ldots, u^{(k-1)}\right)\Big|_{t=0}$$

and so we can write

$$\sum_{k \geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

We call this a *formal power series solution*. Does our solution $u(t)$ agree with this? That is, do we have

$$u(t) = \sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

in a neighbourhood of 0?

Theorem 2.3 (Cauchy-Kovalevskaya for simple ODEs). If $f(u)$ is real analytic in a neighbourhood of u_0, then the series

$$\sum_{k\geq 0} \frac{u^{(k)}(0)}{k!} t^k$$

converges in a neighbourhood of 0 to the unique solution of eq. (2) given by Picard-Lindelöf.

2.2 Real analyticity and majorants

Suppose $f : (-\varepsilon, \varepsilon) \to \mathbb{R}$ is a smooth function. Therefore, $f^{(n)}(0)$ exists for all $n \geq 0$. Does the partial sums

$$\sum_{n\geq 0} \left|f^{(n)}(0)\right| n! x^n$$

for some $|x| \leq \delta$? No, consider the function

$$f(x) = \begin{cases} \exp\left(-\frac{1}{x}\right) & x > 0 \\ 0 & x \leq 0 \end{cases}$$

This is a smooth function, with $f^{(n)}(0) = 0$ for all n.

Definition 2.4 (real analytic)
Let $\mathcal{U} \subseteq \mathbb{R}^n$ be open and $f : \mathcal{U} \to \mathbb{R}$ is *real analytic* (at x_0) if there exists $r > 0$, $f_\alpha \in \mathbb{R}$ such that

$$f(x) = \sum_{\alpha} f_\alpha (x - x_0)^\alpha$$

when $|x - x_0| < r$.

Remark 2.5.

1. That is, f can be written as a convergent power series and
$$f_\alpha = \frac{D^\alpha f(x_0)}{n!}$$
2. Real analyticity is a local property.
3. f is real analytic on an open set U if it is real analytic at each $x_0 \in U$.
4. We will denote the set of real analytic functions on $\mathcal{U}$ by $C^\omega(\mathcal{U})$.
5. If f is C^ω, then f is smooth (e.g. Weierstrass M-test).
6. If f is real analytic, and $\mathcal{U}$ is connected, then f is uniquely determined in $\mathcal{U}$ by its derivatives $D^\alpha f(x)$ at some point $x \in \mathcal{U}$.
7. In particular, f is real analytic if and only if for any compact $K \subseteq U$, there exists C, r such that
$$\sup_{x\in K} |D^\alpha f(x)| \leq C||\alpha|!|r^{|\alpha|}$$

Exercise: Show $f(x) = 1/x$ and $f(x) = \sqrt{x}$ are real analytic for $x > 0$.

Example 2.6

Recall

$$\frac{1}{1-x} = \sum_{k\geq 0} x^k$$

for $|x| < 1$. Let $r > 0$, and consider

$$f(x) = \frac{r}{r-(x_1+\cdots+x_n)} = \frac{1}{1-\frac{x_1+\cdots+x_n}{r}} = \sum_{k\geq 0}\left(\frac{x_1+\cdots+x_n}{r}\right)^k$$

provided $|x_1+\cdots+x_n| \leq \sqrt{n}\left(\sum_j |x_j|^2\right)^{1/2} = \sqrt{n}\|x\| < r$. By the multinomial theorem (sheet 1),

$$f(x) = \sum_{k\geq 0}\frac{1}{r^k}\sum_{|\alpha|=k}\binom{|\alpha|}{\alpha}x^\alpha = \sum_\alpha \frac{|\alpha|!}{\alpha! r^{|\alpha|}}x^\alpha$$

where

$$\binom{|\alpha|}{\alpha} = \frac{|\alpha|!}{\alpha!}$$

and so,

$$f(x) = \sum_\alpha f_\alpha x^\alpha$$

where

$$f_\alpha = \frac{|\alpha|!}{\alpha!}\frac{1}{r^{|\alpha|}}$$

This series is absolutely convergent near zero, since

$$\sum_\alpha \frac{|\alpha!|}{\alpha!}\frac{x^\alpha}{r^{|\alpha|}} = \sum_{k\geq 0}\left(\frac{|x_1|+\cdots+|x_n|}{r}\right)^k < \infty$$

Definition 2.7 (majorise)

Let $f = \sum_\alpha f_\alpha x^\alpha$ and $g = \sum_\alpha g_\alpha x^\alpha$, we say that g *majorises* f, or g *is a majorant of* f, written $g \gg f$, if $g_\alpha \geq |f_\alpha|$ for all α.

For vector valued functions, we require each component to be a majorant.

Lemma 2.8 (properties of majorants).

1. If $g \gg f$, and g converges for $\|x\| < r$, then f converges for $\|x\| < r$.
2. If $f = \sum f_\alpha x^\alpha$ converges for $\|x\| < r$, then for any $s \in (0, r/\sqrt{n})$, there exists a majorant of f which converges for $\|x\| < s/\sqrt{n}$.

Proof. 1. Looking at the partial sums

$$\sum_{|\alpha|\leq k}|f_\alpha x^\alpha| = \sum_{|\alpha|\leq k}|f_\alpha||x_1|^{\alpha_1}\cdots|x_n|^{\alpha_n} \leq \sum_{|\alpha|\leq k}|g_\alpha||x_1|^{\alpha_1}\cdots|x_n|^{\alpha_n} \leq \sum_\alpha |x_1|^{\alpha_1}\cdots|x_n|^{\alpha_n} = g(\tilde{x})$$

where $\tilde{(x)} = (|x_1|,\ldots,|x_n|)$. So $\left\|\tilde{x}\right\| = \|x\|$, and so if $\|x\| < r$, then g converges at $\tilde{x}$. That is, $g(\tilde{x}) < \infty$. Therefore, we have a uniform bound on the partial sum.

2. Let $s \in (0, r/\sqrt{n})$, and set $y = (s,\ldots,s)$. Then $\left\|y\right\| = s\sqrt{n}$, and by assumption,

$$f(y) = \sum_\alpha f_\alpha y^\alpha$$

converges, as $\|y\| = s\sqrt{n} < r$. So there exists a constant c such that $|f_\alpha y^\alpha| \leq c$. Hence

$$|f_\alpha| \leq \frac{C}{|y^\alpha|} = \frac{C}{|y_1|^{\alpha_1} \cdots |y_n|^{\alpha_n}} = \frac{C}{|s|^{|\alpha|}} \leq \frac{C}{|s|^{|\alpha|}} \frac{|\alpha|!}{\alpha!}$$

So we define

$$g(x) = \frac{Cs}{s - (x_1 + \cdots + x_n)}$$

From the above, g majorises f. □

Lecture 3

Proof of theorem 2.3. We will use the method of majorants. Without loss of generality $u_0 = 0$, and for simplicity, we can assume $n = 1$. We need to find the series coefficients. So

$$\dot{u} = f(u)$$

and so $\dot{u}(0) = f(u(0)) = f(0)$, that is, $u_1 = f(0)$. Next,

$$\ddot{u}(t) = f'(u(t))\dot{u}(t)$$

and so $\ddot{u}(0) = f'(0)f(0)$, that is, $u_2 = f'(0)f(0) = \frac{1}{2!}f'(0)f(0)$. Repeating,

$$u^{(3)}(0) = f''(0)f(0)^2 + f'(0)^2 f(0)$$

and so

$$u_3 = \frac{1}{3!}(f''(0)f(0)^2 + f'(0)^2 f(0))$$

Iterating this procedure,

$$u_k = P_k\left(f(0), \ldots, f^{(k-1)}(0)\right)$$

where P_k is a polynomial in k-variables, with nonnegative coefficients. For example,

$$P_1(x) = x$$
$$P_2(x, y) = \frac{1}{2!}xy$$
$$P_3(x, y, z) = \frac{1}{3!}(x^2 z + xy^2)$$

Since f is real analytic, we have that

$$f(v) = \sum_{k \geq 0} f_k v^k$$

where $f_k = \frac{1}{k!}f^{(k)}(0)$. Hence we have that

$$f^{(k)}(0) = k! \cdot f_k$$

Substituting, we have that

$$u_k = Q_k(f_0, \ldots, f_{k-1})$$

which again is a polynomial in k-variables and nonnegative coefficients. This polynomial is "universal".

Aim: We would like to show that the power series

$$\sum_k u_k t^k$$

converges in a neighbourhood of $t = 0$, and solves the ODE, eq. (2). Since f is analytic, we know that

$$f(u) = \sum_k f_k u^k$$

on $|u| < k$. Fixing some $s < r$, there exists a majorant

$$g(u) = \sum_k g_k u^k$$

of f, from lemma 2.8 (ii). Consider the auxilliary differential equation

$$\dot{w}(t) = g(w(t))$$

and $w(0) = 0$. Using the definition of g, we that

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{Cs}{s - w(t)}$$

We get that

$$w = s \pm \sqrt{s^2 - 2Cst}$$

Due to the initial data, we take the $-$ solution. That is,

$$w = s - \sqrt{s^2 - 2Cst}$$

This is real analytic, for $|t| < s/2C$. This tells us that

$$w(t) = \sum_k w_k t^k$$

converges for $|t| < s/2C$. Moreover,

$$w_k = Q_k(g_0, \ldots, g_{k-1})$$

since Q_k is "universal".

Claim 2.9. w majorises u.

By construction, g majorises f, i.e. $g_k \geq |f_k|$ for all k. Moreover, since Q_k has nonnegative coefficients,

$$w_k = Q_k(g_0, \ldots, g_{k-1}) \geq Q_k(|f_0|, \ldots, |f_{k-1}|) \geq |Q_k(f_0, \ldots, f_{k-1})| = |u_k|$$

Hence by lemma 2.8 (i), we know that the series

$$\sum_k u_k t^k$$

converges for $|t| < s/2C$.

To conclude, set

$$u(t) := \sum_{k \geq 0} u_k t^k$$

and we need to check that it solves eq. (2). Both sides are analytic, so suffices to check the derivatives on each side agree to all orders at $t = 0$. □

Remark 2.10.

1. We can extend to systems, where we replace u_k with

   $$u_k^j = Q_k^j(D^{|\alpha|} \mid |\alpha| \leq k)$$

   For w, we can replace $w^j = w^1$ as before.
2. For the non-autonomous case,

   $$u(t) = f(u, t)$$
   $$u(0) = 0$$

   Consider $v(t) = (u(t), t)$, then $\dot{v}(t) = (\dot{u}(t), 1) = (f(u, 1), 1) = (f(v), 1) = F(v)$ with $v(0) = 0$, and we can apply the system version.

For the PDE version, see the handout.

3 Cauchy-Kovalevskaya for PDEs

Let $u : \mathbb{R}^n \to R^m$, and choose $r > 0$. Consider the following problem

$$u_t = \sum_{j=1}^{n} B_j(u, x) u_{x_j} + C(u, x)$$

on $\|x\|^2 + t^2 < r^2$, with

$$u = 0$$

on $\|x\|^2 + t^2 < r^2$ and $t = 0$. The B_j are matrices, B_j and C are real analytic.

Theorem 3.1 (Cauchy-Kovalevskaya for first order systems)**.** Suppose B_j, C are real analytic, for small $r > 0$. Then there exists a unique real analytic function

$$u = \sum_{\alpha} u_\alpha x^\alpha$$

solving the above PDE.

Idea. Compute

$$u_\alpha = \frac{D^\alpha u}{\alpha!}$$

in terms of B_j, C, and show that the power series converges for small r. We use the PDE to find all derivatives. □

Example 3.2

Consider the system

$$u_t = v_x - f$$
$$v_t = -u_x$$

with $u = v = 0$ on $t = 0$. The boundary conditions give us that

$$u(0, 0) = v(0, 0) = 0$$

We would like to determine u_α for all α. By differentiating the boudary conditions,

$$\partial_x^n u(x, 0) = \partial_x^n v(x, 0) = 0$$

for all n. That is, for the case $\alpha = (n, 0)$. From the PDE,

$$u_t(x, 0) = 0 - f = -f \qquad v_t(x, 0) = 0$$

This then means that

$$\partial_x^n \partial_t u(x, 0) = -\partial_x^n f(x, 0)$$

and

$$\partial_x^n \partial_t v(x, 0) = 0$$

for all $n \geq 1$.

Next, if $\alpha = (n, 2)$ use the PDE and we get

$$u_{tt}(x, 0) = f_t(x, 0)$$

and

$$v_{tt}(x, 0) = f_x(x, 0)$$

The same method as above gives us that

$$\partial_x^n \partial_t^2 u(x,0) = -(\partial x)^n \partial_t f(x,0)$$
$$\partial_x^n \partial_t^2 v(x,0) = (\partial_x)^{n+1} f(x,0)$$

Repeating this, we can compute all of the derivatives.

4 Sobolev spaces

5 Elliptic PDEs

6 Hyperbolic PDEs

Index

Cauchy-Kovalevskaya Theorem
    for first order systems, 9
    for simple ODEs, 5
classical solution, 1

data, 3

fully non-linear PDE, 4

linear PDE, 3
    homogeneous, 3

majorise, 6
multi-index notation, 3

PDE of order k, 1
Picard-Lindelöf, 4

quasilinear PDE, 3

real analytic, 5

semilinear PDE, 3
system of PDEs, 1

well-posed, 3